

Immediate Replenishment

(Immediate Replenishment Strategy)

**Microsoft Dynamics 365
Supply Chain Management**

**Advance Warehouse
Management Learning Series**



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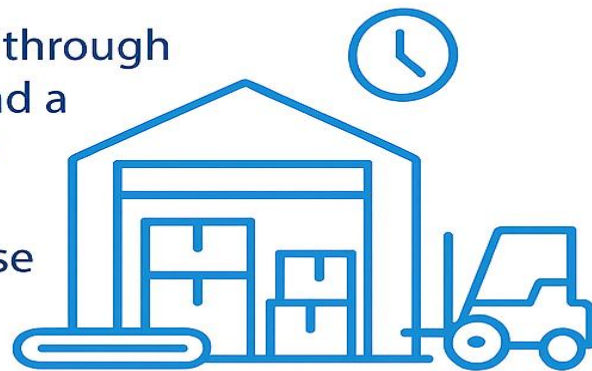


Immediate Replenishment in D365 SCM

“On-the-spot restocking, continuous fulfillment—powered by Immediate Replenishment in D365 SCM”

With Immediate Replenishment in D365 SCM, stock never waits. Replenishment jobs fire instantly, keeping pick locations ready at all times. The system reacts within seconds—cutting delays, boosting availability, and accelerating order fulfillment.

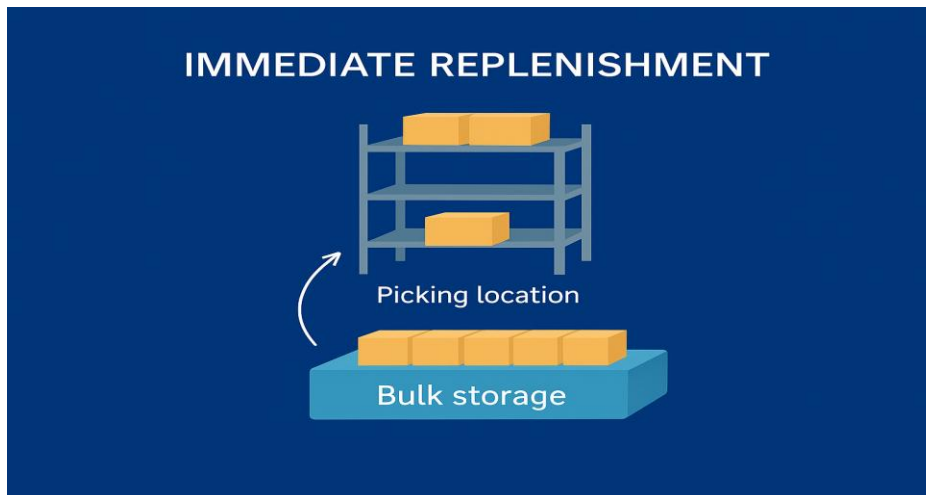
In this article, I’ll walk you through the setup, process flow, and a practical use case to show how Immediate Replenishment can keep warehouse operations flowing and orders continuously fulfilled.





What is Immediate Replenishment?

Immediate replenishment is a process in warehouse management where stock is **refilled instantly as soon as it's needed**. Instead of waiting for a scheduled time or batch (like in wave replenishment), the system makes sure that products are topped up on the spot.



Example:

In a pharmaceutical warehouse, pickers are preparing urgent hospital orders for insulin pens. When the stock at the picking location starts running low, the system triggers immediate replenishment. Fresh cartons are quickly brought from the controlled storage area, ensuring the location is restocked before it runs

Benefits of Immediate Replenishment

1. *Minimized Picking Delays:* Stock is replenished instantly, so pickers don't have to wait.
2. *Improved Order Fulfillment Speed:* Orders are processed faster and shipped on time.
3. *Reduced Manual Intervention:* Automated triggers cut down on human monitoring.
4. *Optimized Inventory Flow:* Ensures smooth movement of goods between reserve and pick locations.
5. *Handles Urgent Demand:* Responds quickly to sudden shortages or spikes in demand.
6. *Improve Customer Service:* Keeps customers satisfied with complete and timely orders.



Immediate Replenishment vs. Wave Demand Replenishment

Aspect	Immediate Replenishment	Wave Demand Replenishment	Result
Trigger	Activated instantly when stock runs low during picking or production.	Activated in batches, based on planned waves of demand.	Stock is always available in Immediate; Wave may risk shortages between cycles.
Timing	Happens in real-time, right when the shortage occurs.	Happens at scheduled intervals, before or during a picking wave.	Immediate = no waiting; Wave = may involve short waiting periods.
Use Case	Best for urgent, high-demand, or time-sensitive operations (e.g., e-commerce, pharma, manufacturing lines).	Best for stable demand and high-volume picking where planning can be done ahead of time.	Immediate ensures continuity; Wave optimizes bulk efficiency.
Benefit	Prevents delays by ensuring stock never runs out during operations.	Increases efficiency by handling replenishment in bulk rather than repeatedly.	Immediate = speed & reliability; Wave = reduced workload.
Risk	Can increase forklift/material handler movements if triggered too often.	Risk of stockouts if demand spikes between replenishment waves.	Immediate = higher labor/traffic; Wave = risk of missed demand.

Set For Up Immediate Replenishment

1. Wave Step Code
2. Wave Template
3. Work template
4. Location Directive
5. Replenishment Template

Wave Step Code

Navigation: Warehouse management > Setup > Waves > Wave Step codes

- Click **New** to create a wave step code.
- Enter the **Wave Step Code** in the respective field.
- Provide a clear **Description** of the step's purpose.
- Select **Replenishment** from the **Wave Step Type** dropdown list.

WAVE STEP CODE:

A **Wave Step Code** is a short label that links a warehouse process step (like replenishment, packing, or labeling) to the correct template. Instead of typing names manually, the system uses predefined codes to ensure consistency and accuracy.



Finance and Operations		
Warehouse management > Setup > Waves > Wave step codes		
Edit + New Delete Options		
Wave step codes		
Standard view ▾		
Filter		
	Wave s... ↑	Wave step description
	Wave step type	
	Default	Default
	Containerization	
	Replen	Replenishment
	Replenishment	



Wave Template

Navigation: Warehouse management > Setup > Waves > Wave templates

- Select the Wave Template type as **Shipping**.
- Click **New** to create a new wave template.
- Enter the **Wave Template Name** and a **Description**.
- In the General Fast-tab, set the **Automate Replenishment Work Release** toggle to **Yes**.
- In the **Methods** FastTab, move the **Replenishment** method into the **Selected methods** box.

The screenshot displays the 'Wave template' configuration window. On the left, a sidebar lists various wave template types, with 'ADW Shipping' selected. The main area is divided into several sections: 'Wave template information' (name, description, sequence, status), 'Warehouse selection' (dropdown for 'ADW'), 'General' (toggle for 'Automate replenishment work release' set to 'Yes'), 'Wave thresholds' (weight, shipment, line thresholds), 'Default values' (attribute dropdowns), and 'Wave execution notifications'. The 'Methods' section at the bottom shows two columns: 'Remaining methods' and 'Selected methods'. The 'Replenishment' method is highlighted in the 'Selected methods' column with a green box.

WAVE TEMPLATE:

A **Wave Template** is a set of instructions that tells the system what to do when a wave is released. It defines the steps that should happen, such as picking items, moving stock from bulk to picking locations, staging items, or shipping them. In short, it helps automate and standardize warehouse processes so that every sales order, transfer order, or replenishment is handled in the same way without extra manual effort.



Work template

Navigation: Warehouse management > Setup > Work > Work Templates

- Select the **Work Order Type** as **Replenishment**.
- Click **New**, then enter the **Work Template Name** and **Description**.
- Click **Save**.
- Under the Work Template Details section, click **New**.
- Select the Work Type as **Pick**.
- Assign the appropriate **Work Class Id** in the work class id field.
- Click **New** again to add another line.
- Select the Work Type as **Put**.
- Assign the appropriate **Work Class Id** in the work class id field.
- Click **Save** to complete the setup.

Work templates

Standard view

Work order type: Replenishment

Sequence	Work template	Work template description	Valid	Automatic	Work pool ID	Work prior
1	Replenishment ADW	Replenishment ADW				

Work Template Details

Line number	Work type	Custom work type	Mandatory	Stop work	Freeze	Directive code	Work class ID	Quality check temp	Anchoring group ID
1.0000000000	Pick						Replenish		
2.0000000000	Put						Replenish		

WORK TEMPLATE:

Work Template defines the sequence of warehouse tasks (such as pick, put, or move) that the system automatically generates to guide warehouse workers during processes like sales order picking, purchase receiving, or replenishment.

- ❖ This template tells the system how to pick items from bulk and put them on the picking shelves.



Location Directive for Replenishment

Navigation: Warehouse management > Setup > Location directive

- Select the Work order type as **Replenishment**.
- Select the warehouse.

Location Directive Fast tab

- Click **New** and enter a Location directive name.
- Under the **Location directive** Fast tab, again select the Work order type as **Pick**.
- In the **Warehouse** Fast tab, choose the desired scope (Warehouse, Warehouse group, or All). In this example, select **Warehouse**.
- Specify the **Site** and **Warehouse** in their respective fields, then click **Save**.

Add Location Directive Line:

- Click **New** under the **Lines** Fast tab.
- Enter the **From quantity** and **To quantity** values.
- Set Locate quantity to **None**, then click **Save**.

Add Location Directive Action:

- Click **New** under the Location directive actions Fast tab.
- Enter a **Name** for the action.
- In the Fixed location usage field, select either **Fixed or Non-fixed**.
- Click **Edit query** to define rules for finding the most suitable location in the warehouse.

Note: To set up the Put location, repeat the same steps as above. Change the **Work Type** from **Pick** to **Put** and update the **Put location** in the **Edit Query** under **Location Directive Action**.

LOCATION DIRECTIVE:

A **Location Directive** is a set of rules that guide the system in deciding where inventory should be stored and from where it should be picked during warehouse operations. These directives are used in processes like receiving, replenishment, and sales order picking to ensure items are placed in the right locations and retrieved from the most efficient spots in the warehouse.



Location Directive for Sales Order

Navigation: Warehouse management > Setup > Location directive

- Select the Work order type as **Sales Order**.
- Select the warehouse.

Location Directive Fast tab

- Click **New** and enter a Location directive name.
- Under the **Location directive** Fast tab, again select the Work order type as **Pick**.
- In the **Warehouse** Fast tab, choose the desired scope (Warehouse, Warehouse group, or All). In this example, select **Warehouse**.
- Specify the **Site** and **Warehouse** in their respective fields, then click **Save**.

Add Location Directive Line:

- Click **New** under the **Lines** Fast tab.
- Enter the **From quantity** and **To quantity** values.
- Set Locate quantity to **None**,
- Select the desired template in the Immediate Replenishment field and click Save to apply.

Add Location Directive Action:

- Click **New** under the Location directive actions Fast tab.
- Enter a **Name** for the action.
- In the Fixed location usage field, select either **Fixed or Non-fixed**.
- Click **Edit query** to define rules for finding the most suitable location in the warehouse.

Location directives

Sequence number	Name
1	ADW Sales Pick

Warehouse selection

Warehouse selection	Site	Warehouse
	1	ADW

Lines

Sequence	From quan...	To quantity	Unit	Locate quantity	Restrict by ...	Round up t...	Locate pac...	Allow split	Immediate replenis...
1	0.00	99,999.00		None					Demand Replish

Location Directive Actions

Sequence	Name	Fixed location usage	Allow neg...	Batch enab...	Strategy
1	ADW Sales Pick	Only fixed locations for th...			None
2	Non-Fixed location	Fixed and non-fixed locati...			None



Note: To set up the Put location, repeat the same steps as above. Change the **Work Type** from **Pick** to **Put** and update the **Put location** in the **Edit Query** under *Location Directive Action*.

Replenishment Template

Note: In this example, the wave demand replenishment strategy is used. Based on business requirements or warehouse policies, other replenishment strategies can also be applied.

Wave demand quantity strategy

When a sales order or transfer order is released to the warehouse, the system looks at the demand (items needed to fulfill the order). If there isn't enough stock in the **picking location**, the **wave demand replenishment process** kicks in. The **quantity strategy** defines how much to replenish into the picking location.

Navigation: Warehouse management > Setup > Replenishment > Replenishment templates

- Click **New**.
- Enter the **Replenishment name** and **Description** in their respective fields.
- Select the **Replenishment type**.
- Select the **Wave step code**.
- Check the box for **Allow wave demand unreserved quantities**.
- Go to the **Replenishment Fast Tab**.
- Click **New**.
- Enter a **Description** in the description field.
- Select the **Replenishment Unit**.
- Select **Request type** in the request type field.
- Choose the **Directive Code** in the Directive Code field.
- Select the **Work Template** in the work template field.
- Choose the replenishment Strategy as **Wave demand quantity**.
- The **Select product** option should be clicked, the desired product can be selected, and then click **OK**.

REPLENISHMENT TEMPLATE:

A **replenishment template** in warehouse management is used to define rules for automatically restocking inventory, especially in picking locations where items are picked to fulfill orders. When stock in a location drops below a certain level, the system triggers the movement of items from bulk storage to the picking area, ensuring availability. This process is part of wave demand replenishment, which helps keep picking locations stocked during high order processing and prevents delays in order fulfillment.



Finance and Operations Warehouse management > Setup > Replenishment > Replenishment templates USMF

Replenishment templates

Standard view

Overview

Replenish template	Description	Replenishment type	Wave step code	Allow wave demand to use unreserved quantities	Use existing immedi...	Cancel repl...
Demand Replenish	Wave demand replenishm...	Wave demand	Replen	✓		
Min-Max Replenish	Min-Max Replenishment	Minimum or maximum				
Min-Max Variant	Min-Max Replenishment t...	Minimum or maximum				

+ New Delete Move up Move down Select products Select product variants Select locations to replenish Select zones to replenish

Replenishment Template Details

Sequence ...	Description	Replenishment unit	Request type	Directive code	Work template	Replenishment strategy	Product query mode
1	Wave demand replenishm...	ea	Wave	Replenish	Rep ADW	Wave demand quantity	Product query

Product query

Select query

Query used Modify...

Range Sorting Joins

+ Add Remove

Table	Derived table	Field	Criteria
Items	Items	Item number	A0001

☐ Replace the criteria value on lookup

OK Reset Cancel



Process Immediate Replenishment strategy

Inventory Overview Before Replenishment

- Location **001-Bulk** contains **30 units** of item **A0001**.
- Location **003-Bulk** contains **20,000 units** of the same item.

The screenshot shows the SAP 'On-hand' inventory overview for item A0001. The interface includes a top navigation bar with 'Finance and Operations', 'Inventory management', 'Inquiries and reports', and 'On-hand list'. Below this is a sub-navigation bar with 'Transactions', 'Warehouse transactions', 'Intercompany on-hand', 'Quantity adjustment', 'Released product details', 'Dimensions', 'View', 'Batch tracking', 'Inspect', 'Related information', and 'Options'. The 'Related information' section is active, showing 'Net requirements', 'Supply overview', 'ATP information', and 'Formula'. The 'On-hand' section is expanded, showing a 'Standard view' with a filter and 'Inventory quantities'. The table below displays the inventory data for item A0001.

Item number	Product name	Search name	Site	Warehouse	Location	Inventory status	License plate	Physical inventory	Physical reserved	Ava
A0001	HDMI 6' Cables	HDMI 6' Cables	1	ADW	0001-Bulk	Available		30,00		
A0001	HDMI 6' Cables	HDMI 6' Cables	1	ADW	0003-Bulk	Available		20,000,00		

Use case for Immediate Replenishment

When a sales order is created for a quantity of 150 units of item **A0001**, the system will first allocate the available **30 units from location 001-Bulk**. Since the demand is higher, the system will then **trigger immediate replenishment from location 003-Bulk (which has 20,000 units available)**. The remaining **120 units will be replenished into 001-Bulk from 003-Bulk**, and the order will be fully fulfilled.



Sales Order:

Finance and Operations | USMF

Actions: Release to warehouse

Sales order details | Standard view

000921 : Contoso Retail San Diego

Lines | Header

Sales order header

Sales order lines

T...	Variant number	Item number	Product name	Sales category	CW quantity	CW unit	Quantity	Unit	Delivery type	CW deliver...	Adjusted u...	Site	Warehouse
		A0001	HDMI 6' Cables				150.00	Pcs	Stock		0.00000	1	ADW

Note: Set inventory reservation to **Automatic** in the Account receivable parameters.

- Under the **Warehouse** action pane, select **Release to warehouse** to release the item.
- Once sales orders is released, the multiple messages will be populated.
 - ✓ Work has been created under work build ID USMF-000000122
 - ✓ Wave USMF-0000000093 has been posted.
 - ✓ Replenishment wave USMF-0000000094 has been created.
 - ✓ 1 shipment has been created.
 - ✓ Work USMF-000084 cannot be unblock because it has unfinished replenishment work.

Finance and Operations | USMF

Work has been created under work build ID USMF-000000122.

Wave USMF-0000000093 has been posted.

Replenishment wave USMF-0000000094 has been created.

Work USMF-000125 cannot be unblocked because it has unfinished replenishment work.

Actions: Release to warehouse

Sales order details | Standard view

000921 : Contoso Retail San Diego

Lines | Header

Sales order header

Sales order lines

T...	Variant number	Item number	Product name	Sales category	CW quantity	CW unit	Quantity	Unit	Delivery type	CW deliver...	Adjusted u...	Site	Warehouse
		A0001	HDMI 6' Cables				150.00	Pcs	Stock		0.00000	1	ADW

- **The system creates two works:** one for replenishment and one for the sales order. Always complete replenishment first, because the sales order work will not continue until replenishment is done.



Finance and Operations Warehouse management > Work > All work USMF

Work Shipment Related Information Options

Maintain Edit Complete work Cancel work Cancel work using batch Work Cycle counting Work exceptions log Change user Work Split work Cancel work split session Block work Unblock work Print Work report

All work

Standard view

Filter Show closed work

Work ID	Work creation num...	Work status	Work order type	Order number	Site	Warehouse	Work in process	Closed work	Target license plate...	Blocked w...	Load ID
USMF-000124	USMF-000000123	Open	Replenishment		1	ADW					
USMF-000125	USMF-000000122	Open	Sales orders	000921	1	ADW				✓	USMF-00008

Navigation: Warehouse management > Work > All Work

Replenishment Work:

Finance and Operations Warehouse management > Work > All work USMF

Work Shipment Related Information Options

Work Edit Work

Standard view

Show closed

Overview General Blocking reasons

Work prior...	Locked by	Work ID	Work status	Site	Warehouse	Work pool ID	Target license plate...	Work order type	Order number	Description	Blocked	Load ID
		USMF-000124	Open	1	ADW			Replenishment				

Lines General

Transactions User actions Custom data Cancel work line Display dimensions Quality check results

Work status	Work type	Location	Item number	Product name	Work quan...	Remaining...	Unit	User ID	Stop trans...	Mandatory	Work class...	Configuration	Size
Open	Pick	0003-Bulk	A0001	HDMI 6' Cables	120.00	120.00	pcs			✓	Replenish		
Open	Put	0001-Bulk	A0001	HDMI 6' Cables	120.00	120.00	pcs			✓	Replenish		



Sales order work

Finance and Operations Warehouse management > Work > All work USMF

Work

Standard view

Show closed

Overview General Blocking reasons

Work prior...	Locked by	Work ID	Work status	Site	Warehouse	Work pool ID	Target license plate...	Work order type	Order number	Description	Blocked	Load ID
Open	30	USMF-000125	Open	1	ADW			Sales orders	000921		✓	USMF-000086

Lines General

Transactions User actions Custom data Cancel work line Display dimensions Quality check results

Work status	Work type	Location	Item number	Product name	Work quan...	Remaining...	Unit	User ID	Stop trans...	Mandatory	Work class...	Configuration	Size
Open	Pick	0001-Bulk	A0001	HDMI 6' Cables	150.00	150.00	pcs			✓	Sales Pick		
Open	Put		A0001	HDMI 6' Cables	150.00	150.00	pcs			✓	Sales Pick		
Open	Pick		A0001	HDMI 6' Cables	150.00	150.00	pcs		✓	✓	Sales Load		
Open	Put		A0001	HDMI 6' Cables	150.00	150.00	pcs			✓	Sales Load		

Note: The overall process remains consistent with the steps outlined in my previous article, Wave Demand Replenishment (Wave Demand Quantity Strategy) part. This ensures continuity and provides a clear connection between replenishment strategies.